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(54) **CARGO HANDLING SYSTEM
REPOSITIONABLE SIDE GUIDE**

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CPC **B64D 9/003** (2013.01)

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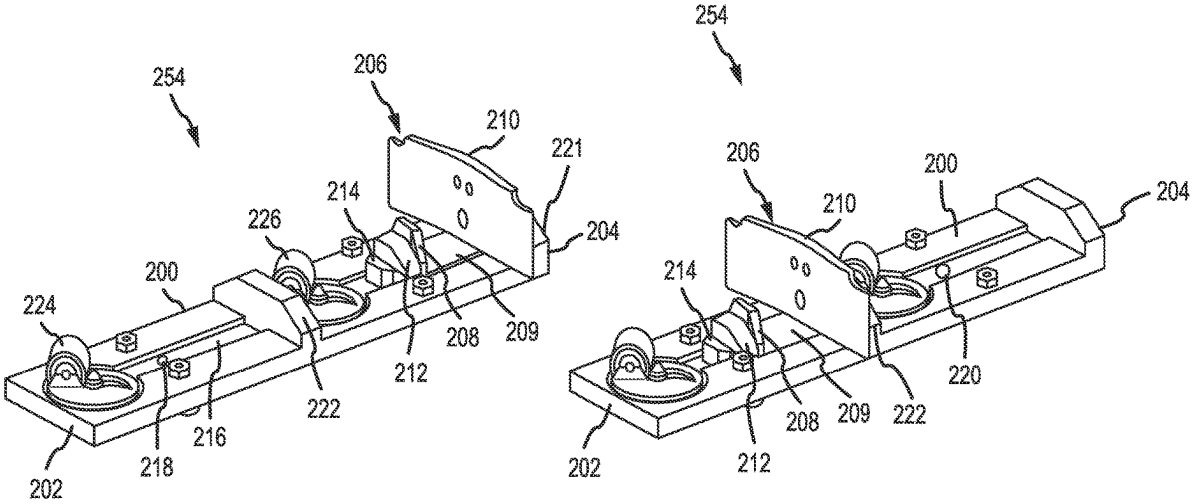
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(57) **ABSTRACT**

A system for repositioning rails in a cargo bay of an aircraft. The system includes a base configured to be coupled to a structure in the cargo bay and having a first side and a second side. The system further includes a cradle for receiving a rail for restricting movement of a unit load device (ULD) in at least one direction, the cradle configured to position the rail between a first position and a second position in which the rail is closer to the second side of the base than when the rail is in the second position.

3 Claims, 5 Drawing Sheets



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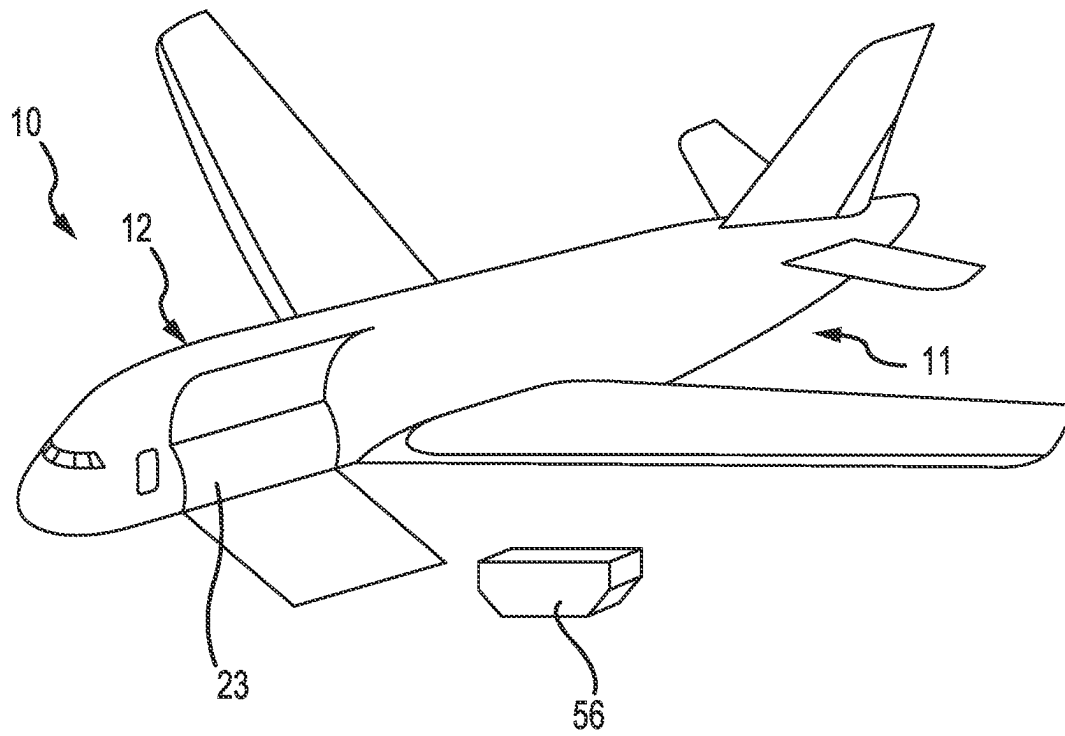


FIG.1

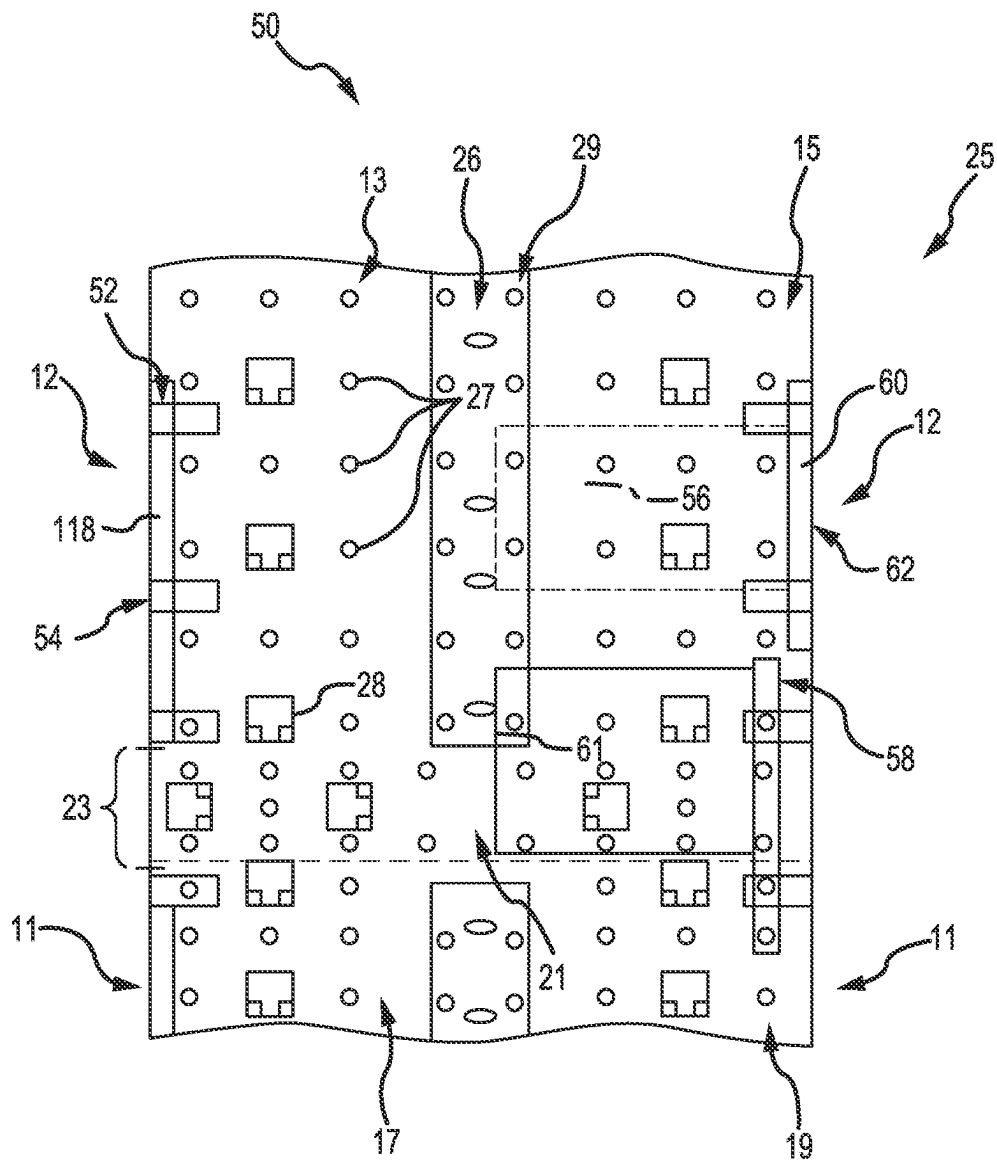


FIG. 2

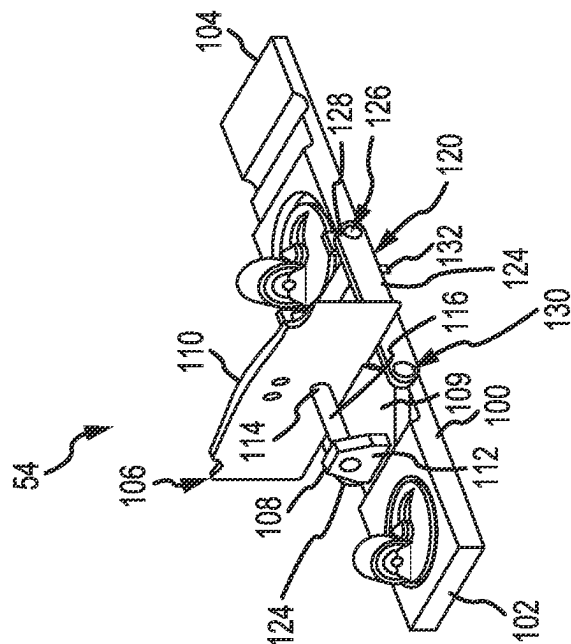


FIG. 3B

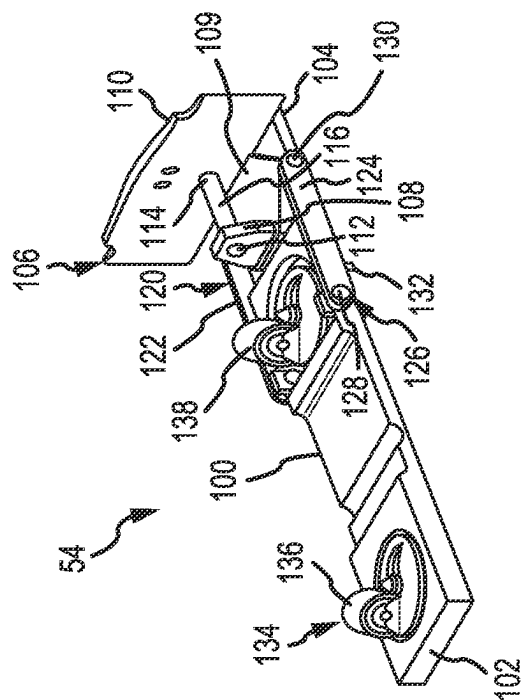


FIG. 3A

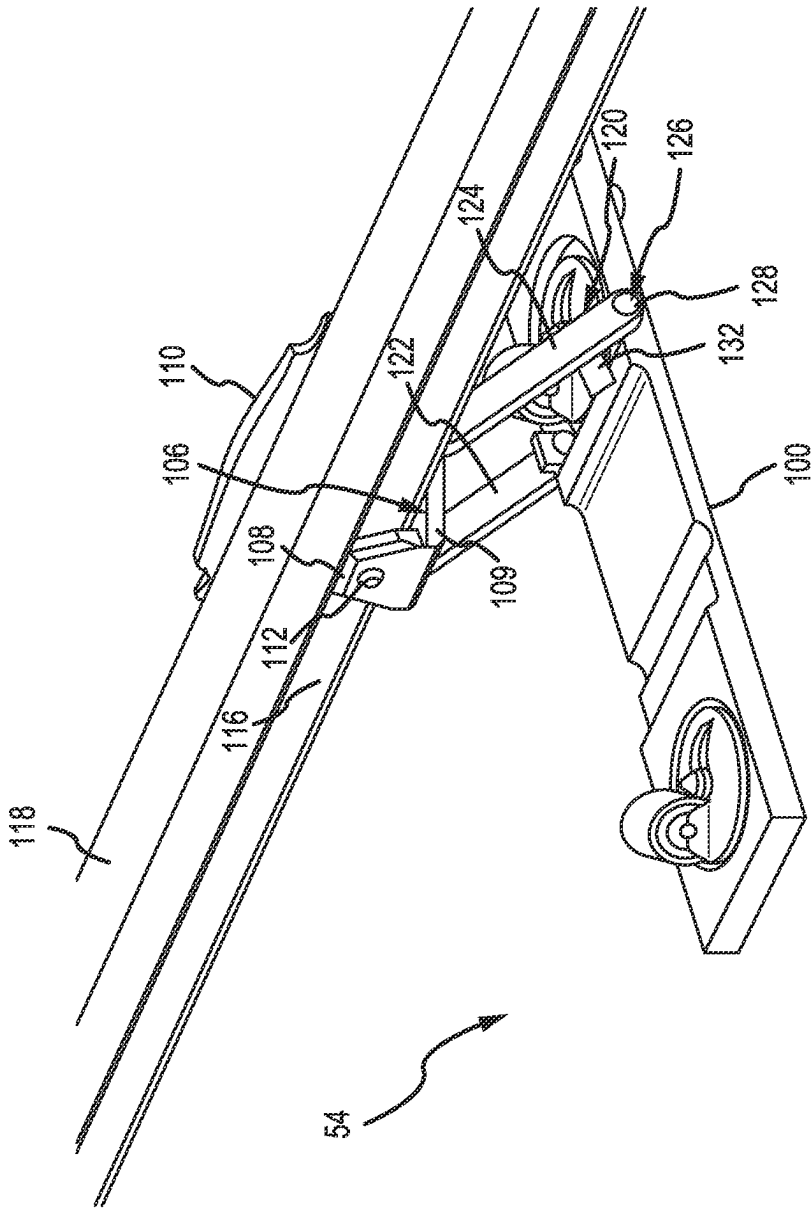


FIG.3C

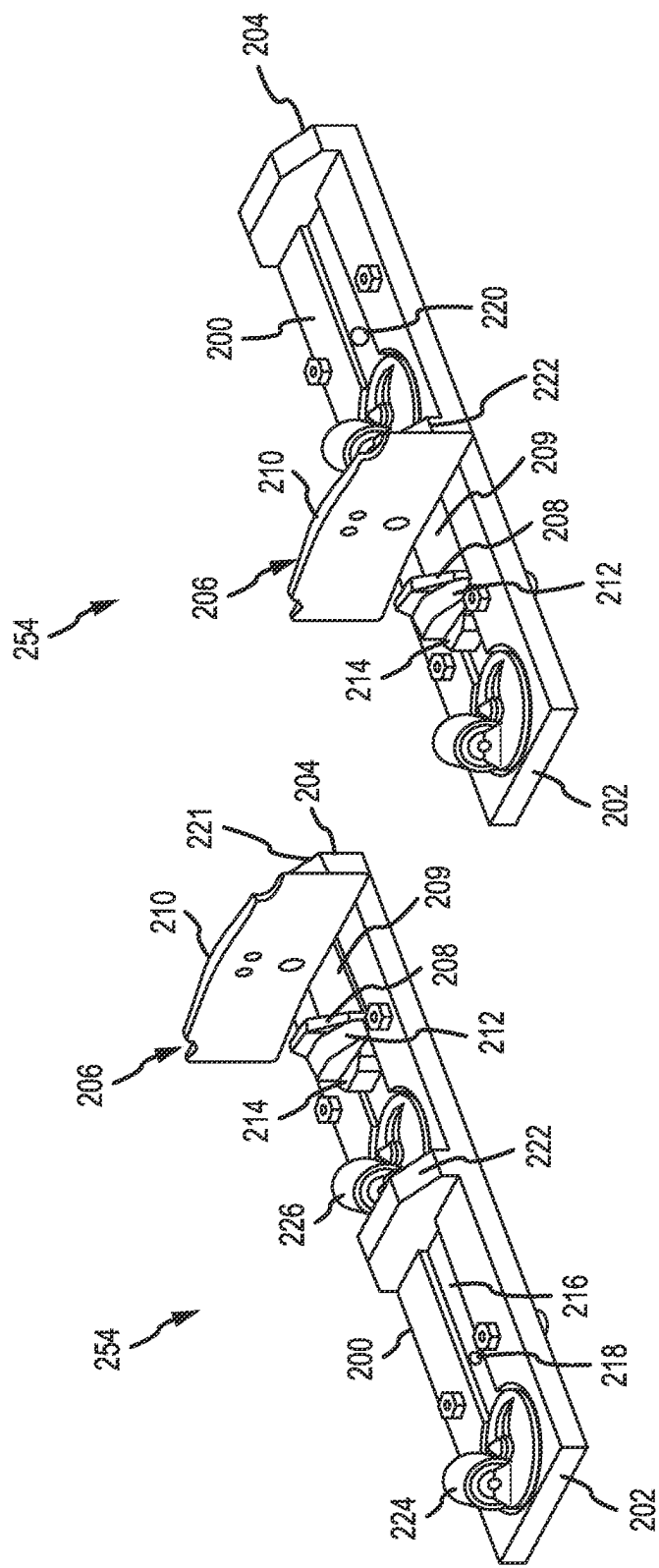


FIG.4B

FIG.4A

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CARGO HANDLING SYSTEM REPOSITIONABLE SIDE GUIDE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of, claims priority to, U.S. application Ser. No. 16/717,511, filed Dec. 17, 2019 and titled "CARGO HANDLING SYSTEM REPOSITIONABLE SIDE GUIDE," which is incorporated by reference herein in its entirety for all purposes.

FIELD

The present disclosure is directed to systems for reconfiguring a cargo bay of an aircraft by adjusting a position of a rail of a cargo system in the cargo bay.

BACKGROUND

Aircraft may transport at least one of passengers or cargo. Many aircraft thus have at least one cargo bay designed to receive cargo. In many aircraft applications, the size of cargo pallets (e.g., unit load devices (ULDs)) is a standardized set of sizes. For example, a first aircraft may be designed to transport ULDs having a first dimension and a second aircraft may be designed to transport ULDs having a second dimension that is different than the first dimension. An aircraft operator who has both aircraft may desire to reconfigure the system one type of aircraft to carry the ULDs from the second type. Alternatively, because of the different sizes of the ULDs, some aircraft are optimized to carry one size. This may result in an inefficient use of space for other ULD sizes.

SUMMARY

Described herein is a system for repositioning rails in a cargo bay of an aircraft. The system includes a base configured to be coupled to a structure in the cargo bay and having a first side and a second side. The system further includes a cradle for receiving a rail for restricting movement of a unit load device (ULD) in at least one direction, the cradle configured to position the rail between a first position and a second position in which the rail is closer to the second side of the base than when the rail is in the second position.

In any of the foregoing embodiments, the cradle includes a first wing extending away from the base and a second wing extending away from the base and located closer to the second side of the base than the first wing.

In any of the foregoing embodiments, the first wing and the second wing each define a fastener aperture such that the rail is configured to be retained in place relative to the cradle by extending a fastener through the fastener aperture of the first wing and the second wing and through the rail.

Any of the foregoing embodiments may further include a plunger configured to be received by the base to resist movement of the plunger, wherein the base further defines a slot configured to receive the plunger and the cradle such that placement of the plunger in the slot resists movement of the cradle.

Any of the foregoing embodiments may further include a support rail coupled to the base between the first side and the second side, wherein the plunger and the cradle are configured to be coupled to the slot on both sides of the support rail.

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Any of the foregoing embodiments may further include at least one support arm coupled to the cradle and pivotably coupled to a pivot point on the base, such that the cradle is configured to move between the first position and the second position by pivoting the at least one support arm about the pivot point on the base.

In any of the foregoing embodiments, the at least one support arm includes two support arms each coupled to the pivot point on the base via a pin.

In any of the foregoing embodiments, the at least one support arm is pivotably coupled to the cradle to facilitate alignment of the cradle relative to the base in both of the first position and the second position.

Any of the foregoing embodiments may further include a locking mechanism configured to lock the cradle in at least one of the first position or the second position.

Any of the foregoing embodiments may further include a roller coupled to the base and configured to facilitate movement of the ULD across the cargo bay.

Also disclosed is a system for repositioning rails in a cargo bay of an aircraft. The system includes a base configured to be coupled to a structure in the cargo bay and having a first side and a second side. The system further includes a roller coupled to the base and configured to facilitate movement of a unit load device (ULD) across the cargo bay. The system further includes a cradle for receiving a rail for restricting movement of the ULD in at least one direction, the cradle configured to position the rail between a first position and a second position in which the rail is closer to the second side of the base than when the rail is in the second position.

Any of the foregoing embodiments may further include a plunger configured to be received by the base to resist movement of the plunger, wherein the base further defines a slot configured to receive the plunger and the cradle such that placement of the plunger in the slot resists movement of the cradle.

Any of the foregoing embodiments may further include a support rail coupled to the base between the first side and the second side, wherein the plunger and the cradle are configured to be coupled to the slot on both sides of the support rail.

Any of the foregoing embodiments may further include at least one support arm coupled to the cradle and pivotably coupled to a pivot point on the base, such that the cradle is configured to move between the first position and the second position by pivoting the at least one support arm about the pivot point on the base.

In any of the foregoing embodiments, the at least one support arm includes two support arms each coupled to the pivot point on the base via a pin.

In any of the foregoing embodiments, the at least one support arm is pivotably coupled to the cradle to facilitate alignment of the cradle relative to the base in both of the first position and the second position.

Also disclosed is a system for repositioning rails in a cargo bay of an aircraft. The system includes a base configured to be coupled to a structure in the cargo bay and having a first side and a second side. The system further includes a cradle for receiving a rail for restricting movement of a unit load device (ULD) in at least one direction, the cradle having a first wing extending away from the base and a second wing extending away from the base and located closer to the second side of the base than the first wing, and the cradle being configured to move the rail between a first

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position and a second position in which the rail is closer to the second side of the base than when the rail is in the second position.

Any of the foregoing embodiments may further include a plunger configured to be received by the base to resist movement of the plunger, wherein the base further defines a slot configured to receive the plunger and the cradle such that placement of the plunger in the slot resists movement of the cradle.

Any of the foregoing embodiments may further include at least one support arm coupled to the cradle and pivotably coupled to a pivot point on the base, such that the cradle is configured to move between the first position and the second position by pivoting the at least one support arm about the pivot point on the base.

In any of the foregoing embodiments, the at least one support arm is pivotably coupled to the cradle to facilitate alignment of the cradle relative to the base in both of the first position and the second position.

The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosures, however, may best be obtained by referring to the detailed description and claims when considered in connection with the drawing figures, wherein like numerals denote like elements.

FIG. 1 illustrates an underside of an aircraft, in accordance with various embodiments;

FIG. 2 illustrates an aircraft cargo deck of the aircraft of FIG. 1, in accordance with various embodiments;

FIG. 3A illustrates a system for adjusting a location of a rail in the aircraft cargo deck of FIG. 2, in accordance with various embodiments;

FIG. 3B illustrates the system of FIG. 3A in a reconfigured position, in accordance with various embodiments;

FIG. 3C illustrates the system of FIG. 3A with a rail attached to a cradle of the system, in accordance with various embodiments;

FIG. 4A illustrates a system for adjusting a location of a rail for use in an aircraft cargo deck, in accordance with various embodiments; and

FIG. 4B illustrates the system of FIG. 4A in a reconfigured position, in accordance with various embodiments.

DETAILED DESCRIPTION

The detailed description of exemplary embodiments herein makes reference to the accompanying drawings, which show exemplary embodiments by way of illustration and their best mode. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that logical, chemical, and mechanical changes may be made without departing from the spirit and scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. For example, the steps recited in any of the method or process descriptions may be executed

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in any order and are not necessarily limited to the order presented. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact.

Referring to FIGS. 1 and 2, an aircraft cargo deck 29 may be used to implement various embodiments of the present disclosure. A conveyance surface 26 forms a deck of an aircraft, adjacent to a cargo deck loading door 23. However, there are many other aircraft cargo deck configurations to which the embodiments of the disclosure can be implemented. For example, various aircraft, particularly those configured primarily for the transportation of cargo without passengers, may lack an upper passenger deck and may utilize an additional cargo deck installed in place of the upper passenger deck. The present disclosure allows for reconfiguration of the cargo deck 29 via the use of systems 52, including a system 54, that provides for repositioning of rails 58 (including a rail 118) in the cargo deck 29.

The cargo compartment includes a system 50 having a plurality of freely rotating conveyance rollers 27 mounted in the cargo deck 29 to define the conveyance plane. Cargo loaded onto the aircraft cargo deck 29 may be moved manually throughout the cargo deck 29 upon the freely rotating conveyance rollers 27. However, it may be desirable to electro-mechanically propel the cargo with minimal or no manual assistance, as some cargo may be relatively heavy and difficult to manipulate. In that regard, the H-shaped cargo surface may include a number of power drive units (PDUs) 28 that provide a mechanism upon which cargo is propelled over the conveyance rollers 27. Each PDU 28 may include one or more actuator that may move cargo relative to the cargo deck 29.

In the longitudinal direction, the conveyance surface 26 may include a left track and a right track along which cargo is to be stowed in parallel columns during flight. In the transverse direction, the cargo deck 29 may also be separated into a tail (or "aft") section 11 and a forward section 12. Thus, the left and right tracks may be divided into four sections, including two forward sections 13, 15 and two aft sections 17, 19. In addition to the four sections, there may be an additional path 21 between both tracks at the cargo deck loading door 23. This additional path 21 may divide the cargo deck 29 between the forward section 12 and aft section 11. This path is used to move cargo into and out the aircraft, and also to transfer cargo between the left and right storage tracks. In various embodiments, the cargo deck 29 may be divided into any quantity of sections, from a single section to any quantity that is physically possible. For example, the systems disclosed herein may operate in a cargo handling system having a single longitudinal lane.

In various embodiments, a piece of cargo (referred to as a "unit load device" (ULD) 56)) may be moved relative to the cargo deck 29. Various ULDs may have differing dimensions such as ULD 56 and 61. In that regard, the systems 52 may be used to adjust the rails 58 and 62 within the cargo deck 29 to facilitate movement and storage of ULDs of various sizes. For example, a rail 62 may resist movement of the ULD 56 in at least one direction and a rail 58 may resist movement of the ULD 61 in at least one direction. As mentioned above, the ULD 56 and the ULD 61 may have different dimensions. The systems 52 may adjust the posi-

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tioning of the rails **58** and **60** to facilitate storage of ULDs of various sizes on the cargo deck **29**.

Referring now to FIGS. **3A** and **3B**, additional details of the system **54** are shown. In particular, the system **54** may include a base **100**. The base **100** may be coupled to a floorboard or other structure in a cargo bay. The base **100** may include a first side **102** and a second side **104**. The second side **104** may be located closer to a wall of the cargo bay than the first side **102**.

The system **54** may further include a cradle **106**. The cradle **106** may be designed to receive a rail (e.g., the rail **118** of FIG. **3C**). The cradle **106** may include a first wing **108** extending outward from the base **100**, a second wing **110** likewise extending outward from the base **100**, and a rail support **109** extending from the first wing **108** to the second wing **110**. The first wing **108** may define a first fastener aperture **112** and the second wing **110** may define a second fastener aperture **114**.

For example and referring to FIGS. **3A**, **3B**, and **3C**, a fastener **116** may extend through the first fastener aperture **112**, the second fastener aperture **114**, and the rail **118** in order to couple the rail **118** to the cradle **106**.

The system **54** may further include support arms **120** which may include a first support arm **122** and a second support arm **124**. The support arms **120** may be pivotably coupled to the base **100** at a pivot point **126** (e.g., via a pivot mechanism). For example, a pin **128** may extend through the support arms **120** and the base **100** in order to pivotably couple the support arms **120** to the base **100**. The support arms **120** may likewise be pivotally coupled to the cradle **106** (e.g., via a pivot mechanism, for example, via pins **130**).

In order to adjust the position of the rail **118** relative to the base **100** (and, thus, the cargo bay), the support arms **120** may pivot about the pivot point **126**, thus moving the cradle **106** (and the rail **118**) between a first position (shown in FIG. **3B**) and a second position in which the cradle **106** is closer to the second side **104** than in the first position (as shown in FIG. **3A**). Due to the pivotable coupling of the support arms **120** to the cradle **106** (i.e., via the pivot mechanism), the cradle **106** (and thus the rail **118**) may remain in an upright (or any configured) position during the transition between the first position and the second position (as shown in FIG. **3C**), and in response to the cradle **106** being in the first position and the second position.

The system **54** may further include a lock, or locking mechanism, **132**. The lock **132** may include any locking mechanism such as a pin and socket, a movable lever (as shown), or the like. As shown, the lock **132** includes a movable lever coupled to at least one of the support arms **120**. The movable lever may slide underneath the base **100** in response to the system **54** being in the desired position (e.g., first position or second position) in order to resist movement of the cradle **106** relative to the base **100**.

In various embodiments, the system **54** may further include rollers **134** (e.g., conveyance rollers) coupled to the base. In particular, the system **54** includes a first roller **136** and a second roller **138**. In various embodiments, the cradle **106** may be positioned on one side (e.g., towards the first side **102**) of the second roller **138** in response to the system **54** being in the second position, and on the other side (e.g., towards the second side **104**) of the second roller **138** in response to the system **54** being in the first position.

Turning to FIGS. **4A** and **4B**, another system **254** may function in a similar manner as the system **54** of FIGS. **3A**, **3B**, and **3C**. In particular, the system **254** may include a base

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200 that has a first side **202** and a second side **204**. The base **200** may be designed to be coupled to a floor panel or other structure in a cargo bay.

The system **254** may further include a cradle **206** having similar features as the cradle **106** of FIGS. **3A**, **3B**, and **3C**. In particular, the cradle **206** includes a first wing **208**, a second wing **210**, and the rail support **209** connecting the first wing **208** to the second wing **210**. A rail may be designed to be coupled to the cradle **206**.

The base **200** may include a first support rail **221** located at or near the second side **204** along with a second support rail **222** located between the first side **202** and the first support rail **221**.

The system **254** may further include a plunger **212** and a plunger stop **214**. The plunger **212** and the plunger stop **214** are designed to be received by a slot **216** defined by the base. The base **200** may further define first plunger apertures **218** and second plunger apertures **220** located closer to the second side **204** than the first plunger apertures **218**. The plunger **212** and the plunger stop **214** are designed to be received by at least one of the slot **216**, the first plunger apertures **218**, or the second plunger apertures **220**. For example, the plunger **212** and the plunger stop **214** may each be designed to be received by one of the first plunger apertures **218** or by one of the second plunger apertures **220**. For example, a nut may be applied to a portion of the plunger **212** after the plunger **212** is inserted into one of the plunger apertures **218**, **220**. As another example, the plunger **212** may press-fit into one or both of the plunger apertures **218**, **220**.

The cradle **206** is designed to be located in a first position (as shown in FIG. **4B**) and a second position (as shown in FIG. **4A**). The cradle **206** is designed to be closer to the second side **204** in response to the cradle **206** being in the second position.

In order to couple the cradle **206** in the first position, the cradle **206** may be positioned adjacent to the second support rail **222**. The plunger **212** and the plunger stop **214** may be received by the first plunger apertures **218**. In that regard, the cradle **206** (and thus any rail) may be restricted from movement along the base **200** by being sandwiched between the plunger **212** and the second support rail **222**.

In order to couple the cradle **206** in the second position, the cradle **206** may be positioned adjacent to the first support rail **221**. The plunger **212** and the plunger stop **214** may be received by the second plunger apertures **220**. In that regard, the cradle **206** may be restricted from movement along the base **200** by being sandwiched between the plunger **212** and the first support rail **221**.

The system **254** may further include a first roller **224** and a second roller **226**. In various embodiments, the cradle **206** may be located on a first side (e.g., closer to the first side **202**) of the second support rail **222** and the second roller **226** in response to the cradle **206** being in the first position. In various embodiments, the cradle may be located on a second side (e.g., closer to the second side **204**) of the second support rail **222** and the second roller **226** in response to the cradle **206** being in the second position.

Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems,

and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment”, “an embodiment”, “an example embodiment”, etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f), unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

What is claimed is:

1. A system for repositioning rails in a cargo bay of an aircraft, the system comprising:

- a base configured to be coupled to a structure in the cargo bay and having a first side and a second side;
- a first support rail coupled to the base;
- a second support rail coupled to the base, the second support rail spaced apart from the first support rail;
- a first roller coupled to the base and configured to facilitate movement of a unit load device (ULD) across the cargo bay;
- a second roller coupled to the base and configured to facilitate movement of the unit load device (ULD) across the cargo bay, the first support rail disposed between the first roller and the second roller, the second roller disposed between the first support rail and the second support rail;
- a cradle for receiving a rail for restricting movement of the unit load device (ULD) in at least one direction, the cradle configured to position the rail between a first position and a second position, wherein the rail is closer to the second side of the base in the first position than when the rail is in the second position;
- a plunger configured to be received by the base to resist movement of the plunger;
- a first rail configuration with the cradle located on the first side, wherein the first rail configuration comprises the cradle positioned adjacent to the second support rail, the cradle restricted from movement along the base by being sandwiched between the plunger and the second support rail; and
- a second rail configuration with the cradle located on the second side, wherein the second rail configuration comprises the cradle positioned adjacent to the first support rail, the cradle restricted from movement along the base by being sandwiched between the plunger and the first support rail.

2. The system of claim 1, wherein the base further defines a slot configured to receive the plunger and the cradle such that placement of the plunger in the slot resists movement of the cradle.

3. The system of claim 1, further comprising the rail configured to restrict movement of the unit load device (ULD) in the at least one direction, wherein:

- the first rail configuration further comprises the rail coupled to the cradle, and
- the second rail configuration further comprises the rail coupled to the cradle.

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